EMPLOYEE ATTRITION ANALYSIS

Data Analytics Project Report

Tools: Python • Pandas • Matplotlib • Seaborn • Jupyter Notebook

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1. Executive Summary

Employee attrition means employees leaving an organization. This project examines employee-related data to understand the overall attrition rate and explore how attrition varies across departments and employee characteristics. Python and its data-analysis libraries are used to inspect, clean, summarize, and visualize the dataset.

Dataset note: Numerical results in this report correspond to the commonly used IBM HR Analytics Employee Attrition dataset with 1,470 records and 35 columns. Compare these details with your downloaded CSV; if your file differs, update the results to match your notebook output.

2. Introduction

Employee attrition can increase recruitment and training costs, affect team productivity, and lead to the loss of organizational knowledge. Exploratory data analysis can help organizations recognize patterns associated with employees leaving. The analysis identifies patterns, not proof that any single factor causes attrition.

3. Problem Statement

The organization needs a clear view of employee attrition. Without analyzing the data, it may be difficult to understand its scale or identify departments and employee groups that may need further attention.

4. Objectives

- Load and understand the dataset.
- Check missing values and duplicate records.
- Calculate employee attrition counts and percentage.
- Compare attrition across departments.
- Explore age and monthly income patterns.
- Create clear charts and provide practical recommendations.

5. Dataset Description

The IBM HR Analytics Employee Attrition dataset contains fields such as Age, Department, JobRole, MonthlyIncome, JobSatisfaction, YearsAtCompany, and Attrition. The Attrition field indicates whether an employee left (Yes) or stayed (No).

Dataset item	Description
Dataset	IBM HR Analytics Employee Attrition
Rows	1,470 (standard version)
Columns	35 (standard version)
Target field	Attrition: Yes / No
Analysis type	Exploratory data analysis (EDA)

6. Tools and Technologies

Tool / Library	Purpose
Python	Main programming language
Pandas	Loading, cleaning, filtering, summarizing data

Tool / Library	Purpose
Matplotlib	Creating charts
Seaborn	Statistical visualizations
Jupyter Notebook / VS Code	Writing and running code

7. Methodology

- Import required Python libraries.
- Load the CSV file into a Pandas DataFrame.
- Inspect rows, dimensions, column names, and data types.
- Check missing values and duplicate records.
- Calculate overall attrition counts and rate.
- Compare attrition by department.
- Explore age and monthly income patterns.
- Create charts, review results, and save cleaned data.

8. Data Cleaning and Preparation

Inspect the dataset before analysis to confirm that it loaded correctly and expected columns are available. Check duplicate records and missing values. Choose cleaning operations based on the actual data; do not remove rows automatically without a clear reason. Review data types so numeric and categorical fields can be analyzed correctly.

- data.head()
- data.shape and data.info()
- data.isnull().sum()
- data.duplicated().sum()
- value_counts() for Attrition

9. Data Analysis and Results

9.1 Overall Attrition

In the standard dataset, 237 employees left and 1233 stayed out of 1470 records. The overall attrition rate is 16.12%.

Attrition status	Count	Percentage
Yes — left	237	16.12%
No — stayed	1233	83.88%
Total	1470	100.00%

9.2 Department-wise Attrition

In the standard dataset, Sales has the highest attrition rate among the three departments, followed by Human Resources and Research & Development. Since department sizes differ, rates are more useful for comparison than counts alone.

Department	Employees	Left	Rate
Human Resources	63	12	19.05%
Research & Development	961	133	13.84%
Sales	446	92	20.63%

9.3 Age and Attrition

In the standard dataset, employees who left tend to be younger on average than employees who stayed. This is a sample pattern and does not show that age causes attrition.

9.4 Monthly Income and Attrition

A boxplot can compare monthly income distributions for employees who left and stayed, including medians, spread, and potential outliers. Interpret income alongside job role, level, and experience.

10. Visualizations

Chart	Purpose
Attrition count chart	Shows employees who left versus stayed.
Department-wise attrition chart	Compares attrition across departments.
Age boxplot	Compares age distributions by attrition.
Monthly income boxplot	Compares income distributions by attrition.

Insert actual charts exported from your notebook if required. This report does not include fabricated chart images; visuals should match your own CSV and code.

11. Key Findings

- The standard dataset contains 1470 employee records, with 237 employees marked as having left.
- The overall attrition rate is approximately 16.12%.
- Sales has the highest department-level attrition rate in the standard dataset.
- Employees who left tend to be younger on average in the standard dataset.
- These are descriptive patterns and do not establish cause and effect.

12. Recommendations

- Review employee feedback and exit-interview comments.
- Investigate Sales department attrition, including workload, manager support, and career paths.
- Review job satisfaction, work-life balance, overtime, compensation, and promotion opportunities together.
- Monitor attrition metrics over time.
- Protect employee privacy and support fair, evidence-based decisions.

13. Limitations

- The dataset may not represent every organization.
- Exploratory analysis identifies associations, not causal relationships.
- Results depend on data completeness and accuracy.

- Validate all figures against your actual CSV and notebook output before final submission.

14. Conclusion

This project demonstrates how Python can analyze employee attrition data, check data quality, calculate summary metrics, compare departments, and visualize age and monthly income patterns. A future version could include job satisfaction, overtime, years at company, and job role, followed by a carefully validated predictive model.

15. Appendix: Example Python Workflow

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

data = pd.read_csv("WA_Fn-UseC_-HR-Employee-Attrition.csv")

print(data.head())
print(data.shape)
print(data.info())
print(data.isnull().sum())
print("Duplicates:", data.duplicated().sum())

print(data["Attrition"].value_counts())
print(data["Attrition"].value_counts(normalize=True) * 100)

sns.countplot(data=data, x="Attrition")
plt.title("Employee Attrition Count")
plt.show()

sns.countplot(data=data, x="Department", hue="Attrition")
plt.title("Attrition by Department")
plt.xticks(rotation=15)
plt.tight_layout()
plt.show()

sns.boxplot(data=data, x="Attrition", y="Age")
plt.title("Age by Attrition")
plt.show()

sns.boxplot(data=data, x="Attrition", y="MonthlyIncome")
plt.title("Monthly Income by Attrition")
plt.show()
```

Replace the CSV filename with the exact filename on your computer. If you use a different dataset version, update the code and reported figures accordingly.